COMPUTER NETWORKS - UE24CS252B

Lab 2: Understanding Persistent and Non-persistent HTTP Connections

SRN: PES1UG24CS493

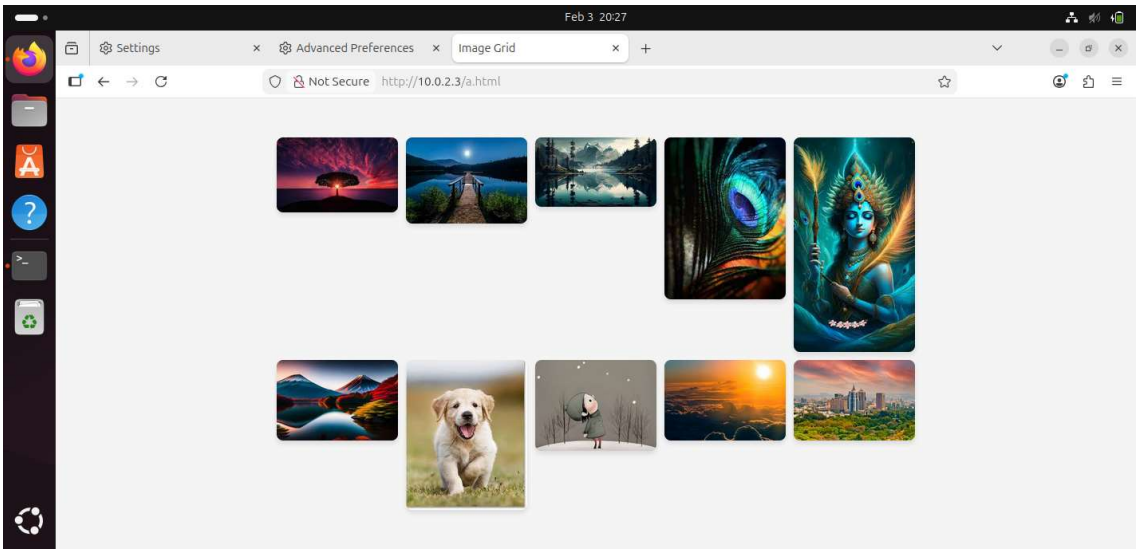
NAME: Tanmayi Nagabhairava

SEC: I

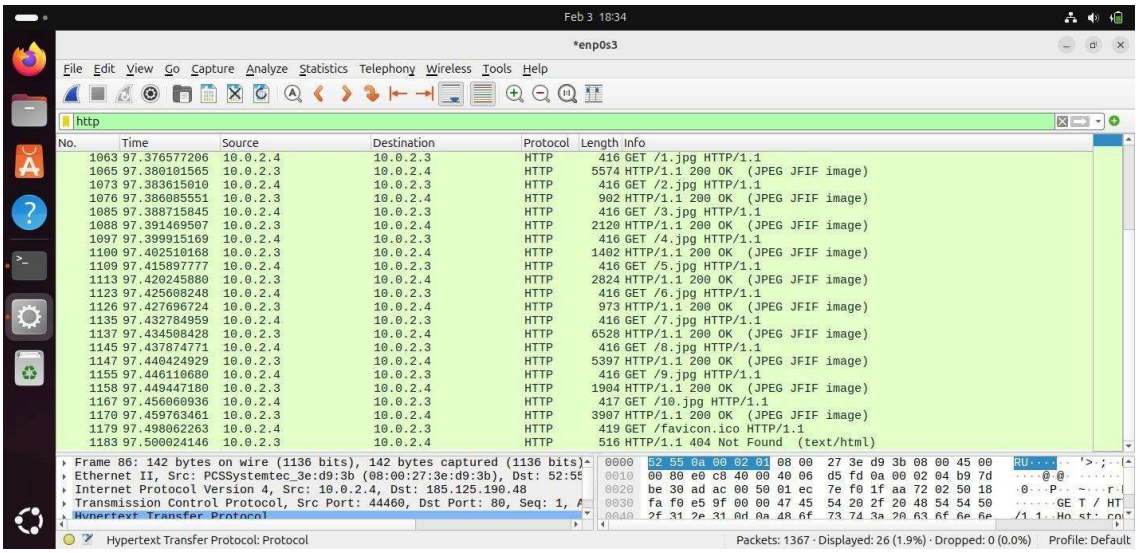
HTTP Connection

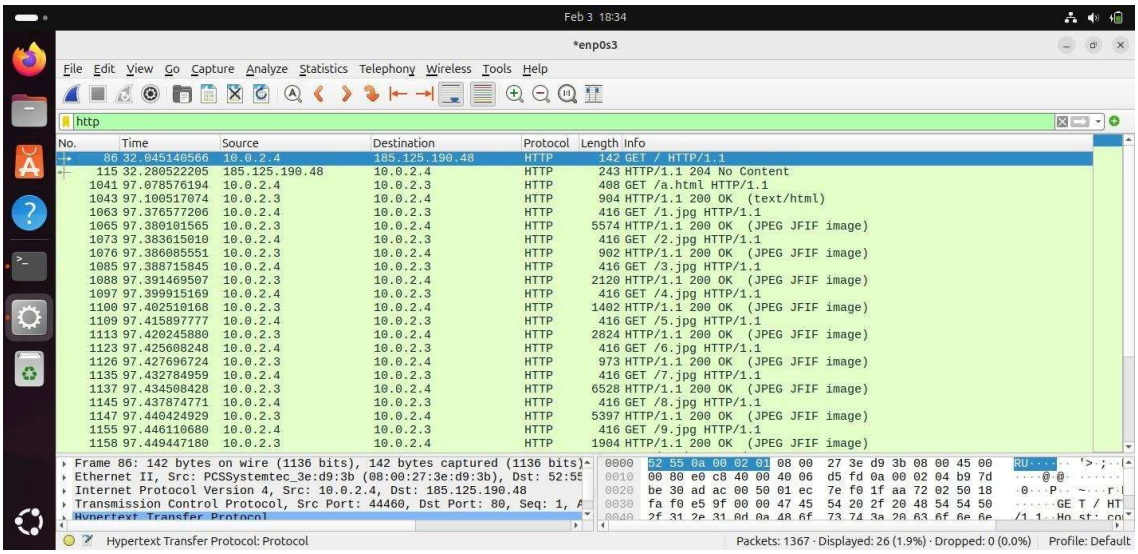
Output:





Non Persistent Connections





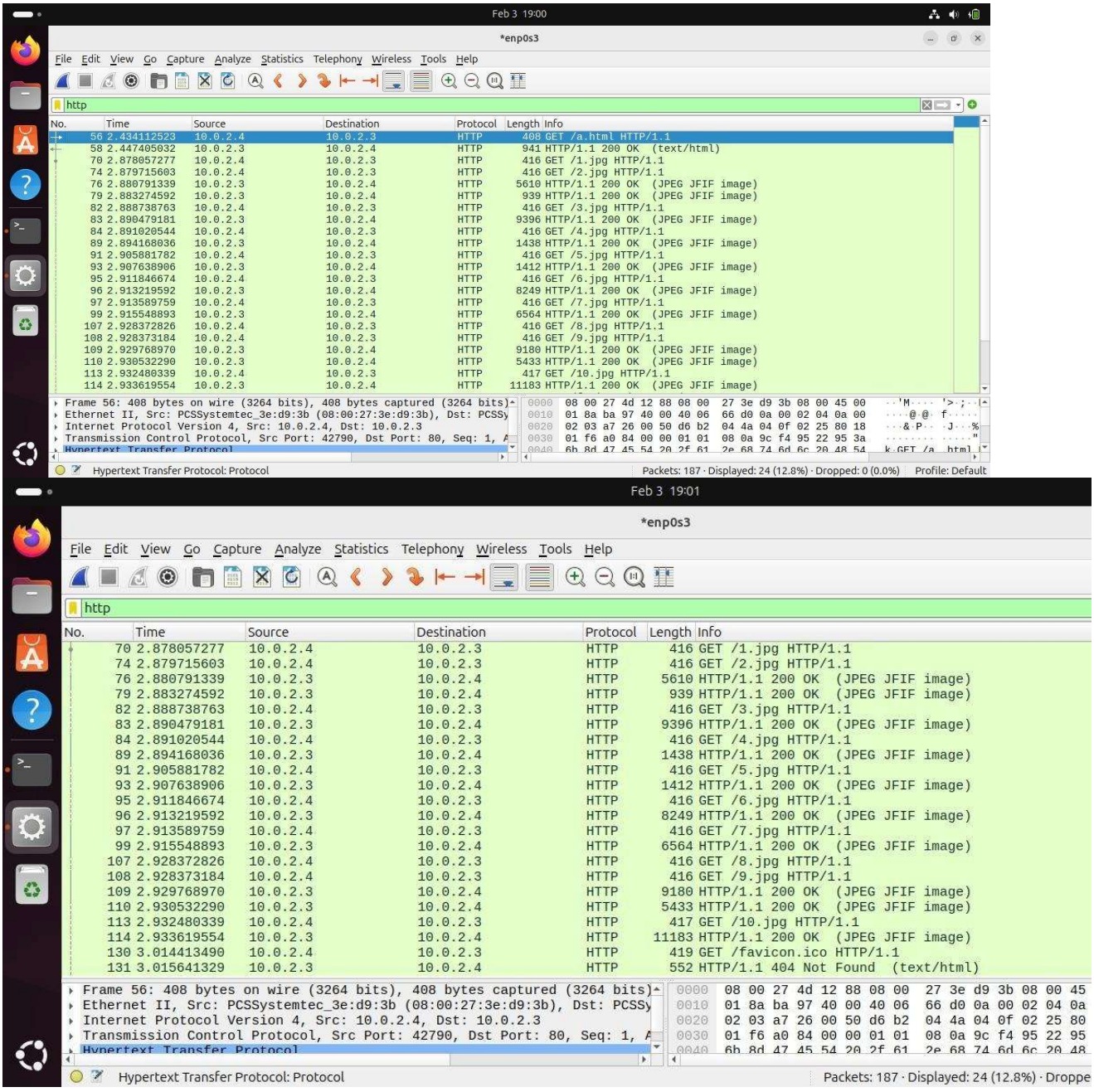
Start Time: 97.078576

End Time: 97.500024

Load Time: 97.500024 - 97.078576 = 0.421448 seconds

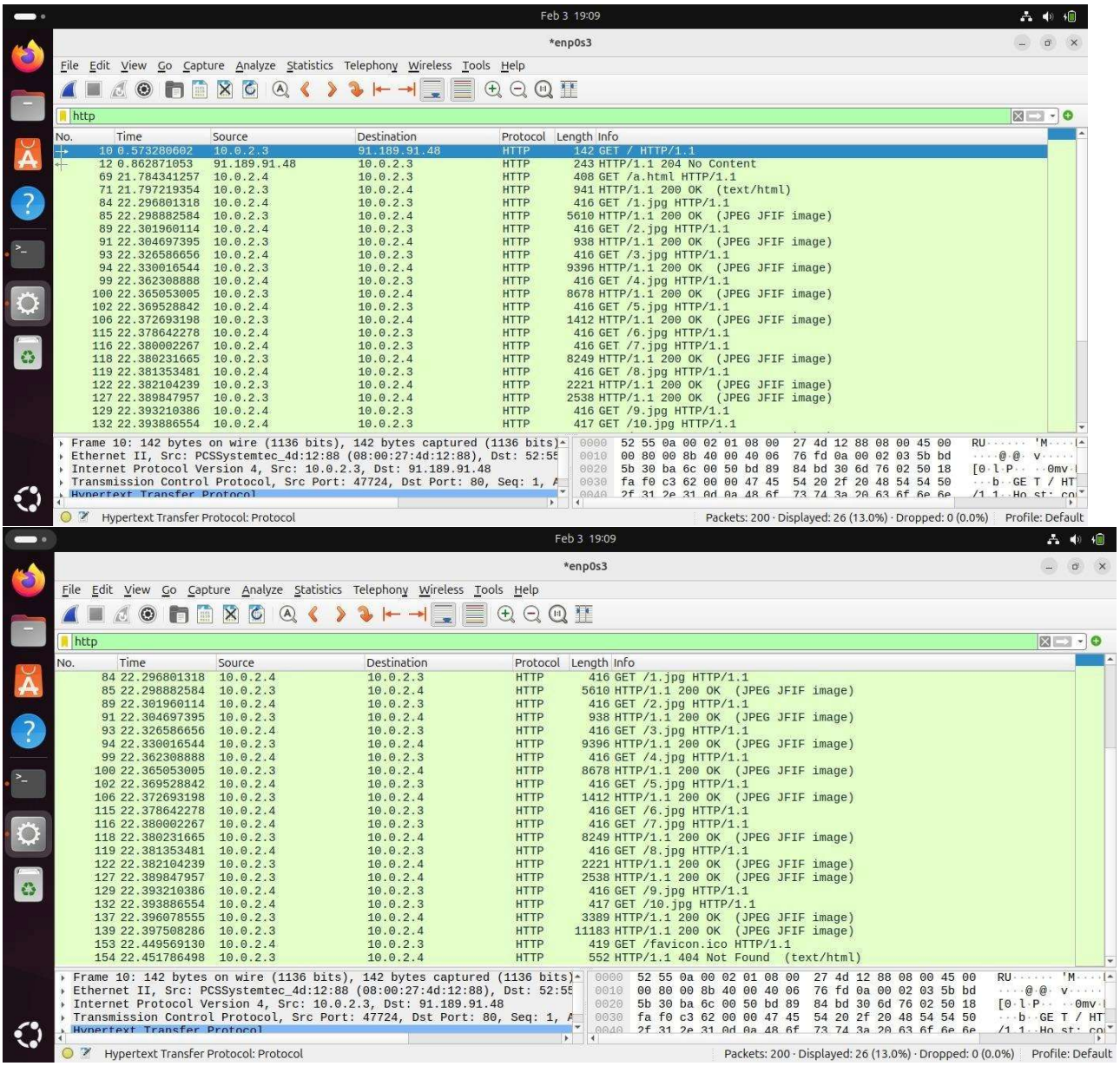
Persistent Connections

Case A: 2 Persistent Connections



Start Time: 2.434112
End Time: 3.015641
Load Time: 3.015641 - 2.434112 = 0.581529 seconds

Case B: 4 Persistent Connections

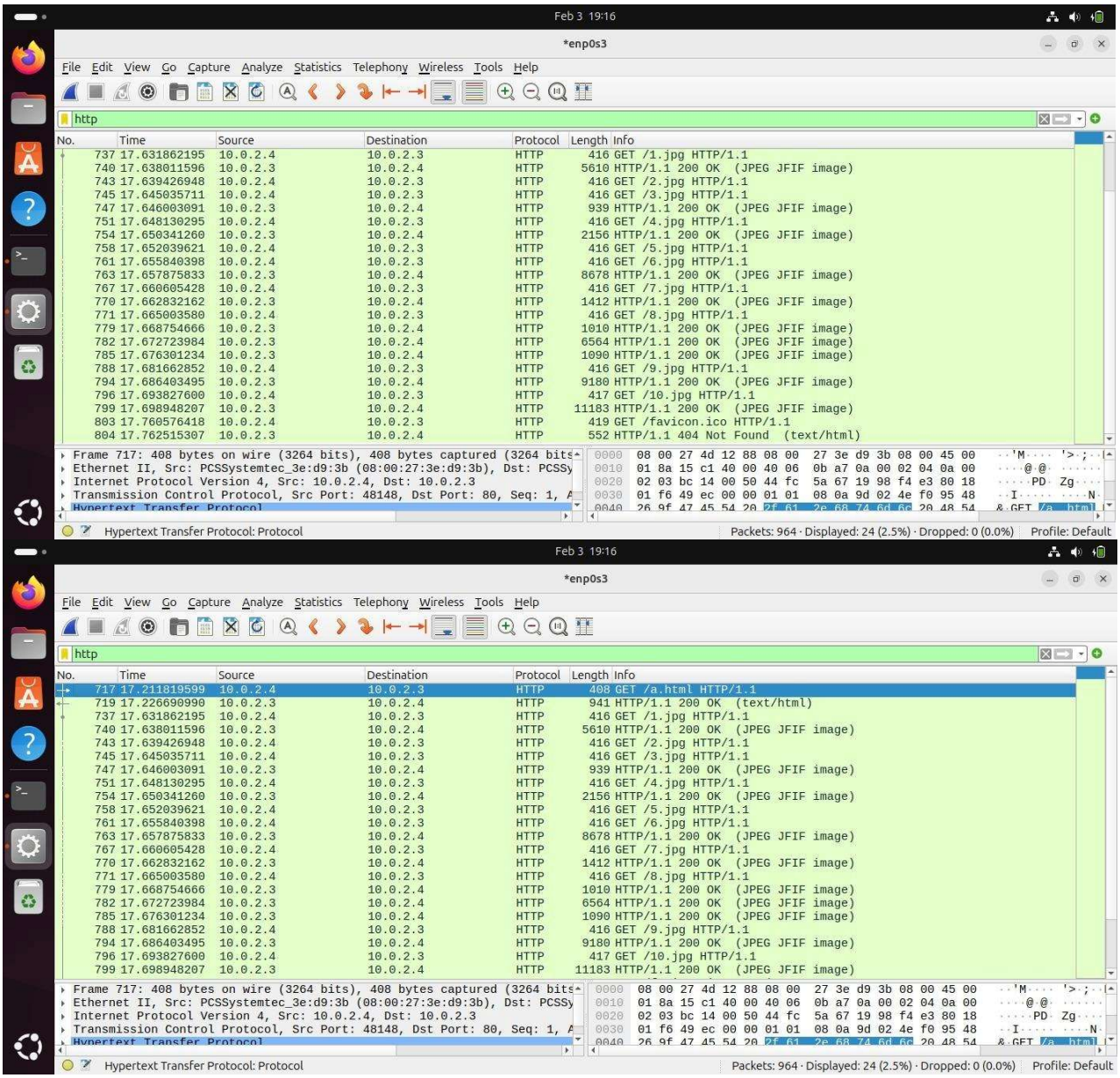


Start Time: 21.784341

End Time: 22.451786

Load Time: 22.451786 - 21.784341 = 0.667445 seconds

Case C: 6 Persistent Connections

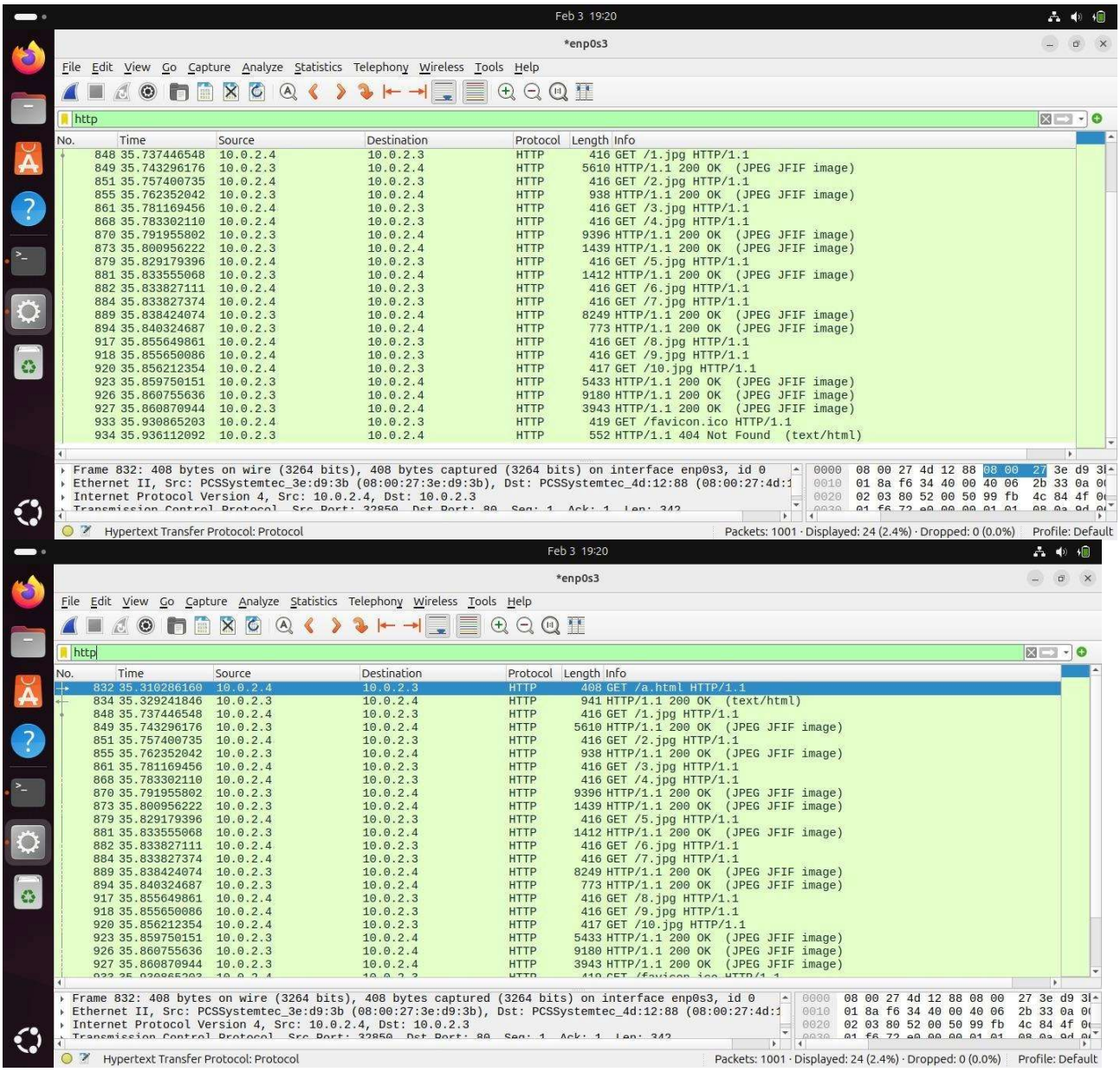


Start Time: 17.211819

End Time: 17.762515

Load Time: 17.762515 - 17.211819 = 0.550696 seconds

Case D: 8 Persistent Connections

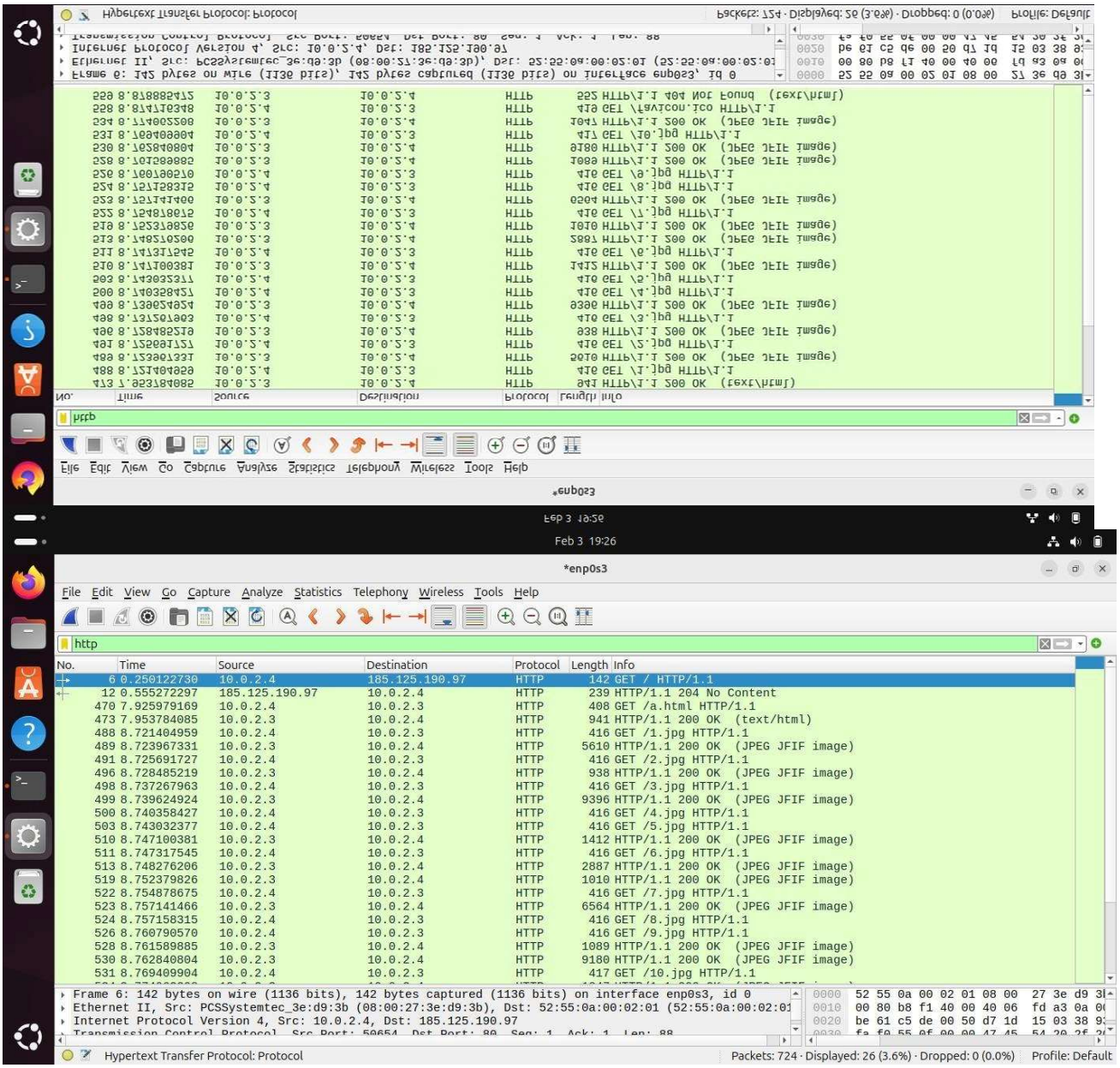


Start Time: 35.310286

End Time: 35.936112

Load Time: 35.936112 - 35.310286 = 0.625826 seconds

Case E: 10 Persistent Connections

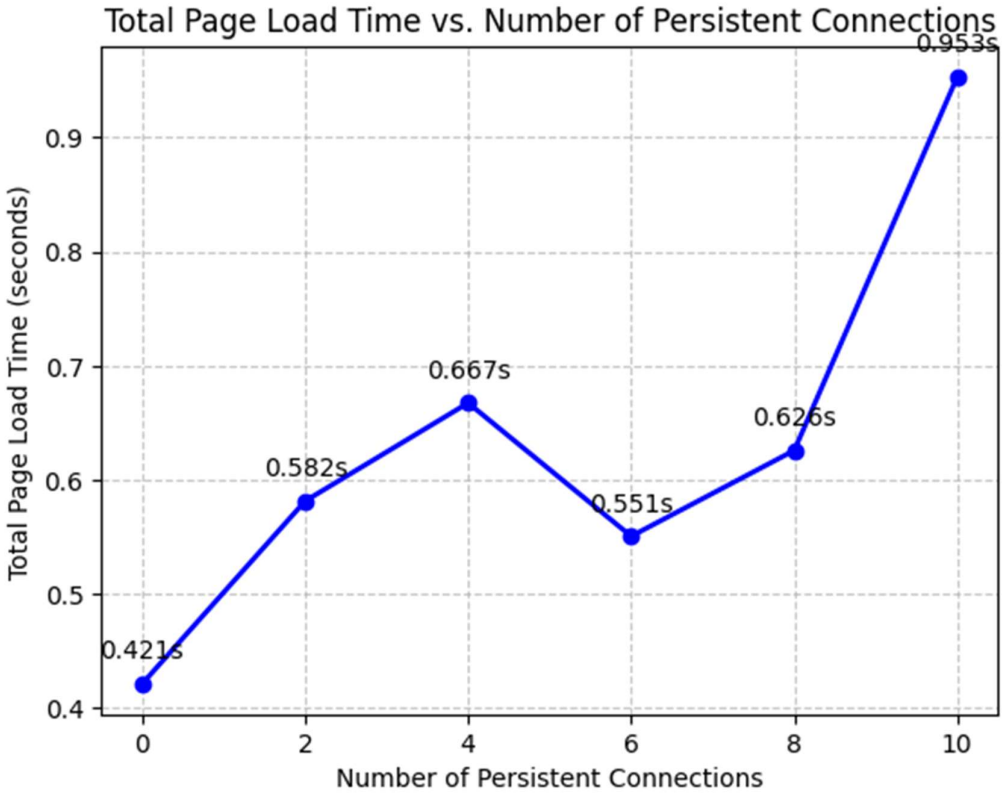


Start Time: 7.925979

End Time: 8.878885

Load Time: 8.878885 - 7.925979 = 0.952906 seconds

Graph:



Conclusion

HTTP connection persistence significantly affects page load time and network efficiency. In non-persistent HTTP, a new TCP connection is created for every object, leading to repeated handshakes, higher protocol overhead, and inefficient use of network resources. Persistent HTTP reduces this overhead by reusing existing TCP connections for multiple requests, which lowers latency and improves bandwidth utilization. This generally results in faster page loading and better efficiency. However, allowing too many simultaneous persistent connections can introduce processing overhead and network congestion, especially on limited systems, which may reduce performance gains. Therefore, HTTP persistence improves page load time and efficiency up to an optimal number of connections, beyond which the benefits may diminish.